

Human Review: Interior APEPs in ℓ_p -sums

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed characterization appears correct and quite natural. The source paper gives the general criterion

$$x \in \text{ape}(C) \iff x \in \overline{\text{ext}(C^{w*})}^{w*},$$

and Proposition 4.9 characterizes the norm-one APEPs in ℓ_p -sums. Question 8.2 asks for the remaining interior case. This packet supplies a full characterization by introducing the admissible coordinate radii.

Main Idea

For each coordinate, set

$$C_i = \overline{\text{ext } B_{X_i^{**}}}^{w*}.$$

For $u \in X_i$, a radius $r \in [0, 1]$ is admissible if

$$u \in rC_i.$$

The least admissible radius $\rho_i(u)$ measures how much p -mass is needed in the i -th coordinate to realize u as a weak-star limit of scaled coordinate extreme points.

With this notation, the characterization is very natural. In the infinite summand case, $x = (x_i)$ is an interior APEP precisely when

$$\sum_i \rho_i(x_i)^p \leq 1.$$

The remaining p -mass can be placed in a coordinate outside any finite weak-star test. In the finite summand case, there is no outside coordinate, so one needs admissible radii $r_i \in R_i(x_i)$ with exact equality

$$\sum_i r_i^p = 1.$$

Proof Check

The necessity argument is sound. If x is an APEP, Theorem 3.8 gives a net of extreme points

$$y^\alpha = (y_i^\alpha) \in \text{ext } B_{X^{**}}$$

weak-star converging to x . Coordinatewise weak-star convergence follows from the coordinate embeddings $X_i^* \hookrightarrow X^*$. For each finite set of coordinates, one passes to a subnet so that

$$\|y_i^\alpha\| \rightarrow r_i.$$

If $r_i = 0$, then $x_i = 0$, hence $0 \in R_i(x_i)$. If $r_i > 0$, then

$$\frac{y_i^\alpha}{\|y_i^\alpha\|} \in \text{ext } B_{X_i^{**}},$$

and after passing to a further subnet one obtains $x_i = r_i e_i$ with $e_i \in C_i$. Thus $r_i \in R_i(x_i)$ and $\rho_i(x_i) \leq r_i$. Since every y^α has norm one, finite partial sums satisfy

$$\sum_{i \in F} \rho_i(x_i)^p \leq 1.$$

Taking the supremum over finite F gives the required inequality in the infinite case, while taking F to be the finite support gives the equality condition in the finite case.

The sufficiency argument is also sound. In the finite case, admissible radii with p -sum equal to one allow one to approximate each coordinate by genuine coordinate extreme points and thereby build global extreme points of $B_{X^{**}}$ weak-star close to x . In the infinite case, finite-support approximation in

$$X^* = \left(\bigoplus_i X_i^* \right)_q$$

reduces a weak-star neighbourhood to finitely many coordinates. One realizes the required coordinates using the least admissible radii, and if there is leftover mass, places it in an unused coordinate. The extreme-point formula for ℓ_p -sums then gives an extreme point of $B_{X^{**}}$ inside the chosen neighbourhood.

Review Outcome

Approve this packet as an apparently correct full solution to Question 8.2. The characterization is not simply copied from the paper: the paper supplies the general weak-star closure criterion and the norm-one ℓ_p -sum case, while this argument extends the analysis to interior points through the admissible-radius formulation.